

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/5882556>

Muscle K⁺, Na⁺, and Cl⁻ disturbances and Na⁺-K⁺ pump inactivation: Implications for fatigue

Article in *Journal of Applied Physiology* · February 2008

DOI: 10.1152/japplphysiol.01037.2007 · Source: PubMed

CITATIONS

178

READS

220

3 authors, including:



[Michael J McKenna](#)

Victoria University Melbourne

140 PUBLICATIONS 6,130 CITATIONS

[SEE PROFILE](#)



[Jean-Marc Renaud](#)

University of Ottawa

79 PUBLICATIONS 2,316 CITATIONS

[SEE PROFILE](#)

Some of the authors of this publication are also working on these related projects:



Effects of post exercise potassium on the heart [View project](#)



Glucose Transporters and Memory [View project](#)

MUSCLE K^+ , Na^+ , Cl^- DISTURBANCES AND Na^+ , K^+ -PUMP INACTIVATION: IMPLICATIONS FOR FATIGUE

by

Michael J. McKenna¹, Jens Bangsbo² and Jean-Marc Renaud³

¹Muscle Ions and Exercise Group, School of Human Movement, Recreation and Performance,
Centre for Ageing, Rehabilitation, Exercise and Sport, Victoria University, Melbourne, Victoria,
Australia,

²Copenhagen Muscle Research Centre, Institute for Sport and Exercise Sciences, University of
Copenhagen, Copenhagen, Denmark,

³University of Ottawa, Department of Cellular and Molecular Medicine, 451 Smyth Rd., Ottawa,
Ontario, Canada, K1H 8M5

RUNNING TITLE: Na^+ , K^+ , Cl^- and Na^+ , K^+ -pumps in muscle fatigue

CORRESPONDENCE: Prof. Michael J. McKenna
School of Human Movement, Recreation and Performance
Centre for Ageing, Rehabilitation, Exercise and Sport, Victoria
University
PO Box 14428, Melbourne, 8001
Victoria, Australia
Tel.: 61-3-9919-4499
Fax: 61-3-9919-4891
e-mail: michael.mckenna@vu.edu.au

ABSTRACT

Membrane excitability is a critical regulatory step in skeletal muscle contraction and is modulated by local ionic concentrations, conductances, ion transporter activities, temperature and humoral factors. Intense fatiguing contractions induce cellular K^+ efflux, Na^+ and Cl^- influx, causing pronounced perturbations in extracellular (interstitial) and intracellular K^+ and Na^+ concentrations. Muscle interstitial K^+ concentration may increase 1 to 2-fold to 11-13 mM and intracellular K^+ concentration fall by 1.3 to 1.7-fold; interstitial Na^+ concentration may decline by 10 mM and intracellular Na^+ concentration rise by 1.5 to 2.0-fold. Muscle Cl^- concentration changes reported with muscle contractions are less consistent, with reports of both unchanged and increased intracellular Cl^- concentrations, depending upon contraction type and the muscles studied. When considered together, these ionic changes depolarize sarcolemmal and t-tubular membranes to depress tetanic force and are thus likely to contribute to fatigue. Interestingly, less severe local ionic changes can also augment sub-tetanic force, suggesting they may potentiate muscle contractility early in exercise. Increased Na^+,K^+ -ATPase activity during exercise stabilize Na^+ and K^+ concentration gradients and membrane excitability and thus protects against fatigue. However, during intense contraction some Na^+,K^+ -pumps are inactivated and together with further ionic disturbances, likely precipitate muscle fatigue.

INTRODUCTION

Muscle contraction requires the propagation of action potentials (AP) along the sarcolemma and down the transverse-tubules where they activate voltage sensors and enable Ca^{2+} release from the sarcoplasmic reticulum. Each AP comprises Na^+ influx during the depolarization phase and K^+ efflux for the repolarization phase. Cl^- also diffuses into the sarcoplasm contributing to the repolarization phase because Cl^- channels remain open during depolarization (9) and the Cl^- equilibrium potential (E_{Cl}) is near the resting membrane potential (E_m) (23; 40). Thus, it is unremarkable that intense muscle contractions also induce pronounced perturbations in muscle Na^+ , K^+ , and Cl^- ions. The important question is whether these changes are linked to fatigue.

Muscle fatigue can be defined as a transient and recoverable decline in muscle force and/or power during repeated or continuous muscle contractions. Whilst the mechanisms of muscle fatigue have been studied extensively, considerable debate and controversy still exists. This is likely in part because muscle fatigue appears to be a multifactorial mechanism, evidenced by other mini-reviews in this series. This review focuses on muscle K^+ , Na^+ and Cl^- disturbances, on Na^+ , K^+ -ATPase (NKA, Na^+ , K^+ -pump) activity and membrane inexcitability as critical components in muscle fatigue. We firstly identify the muscle intracellular and extracellular K^+ , Na^+ and Cl^- concentration changes during contractions; examine their impact on membrane potential and excitability, and thus, their implications for muscle fatigue. We then review the impact of NKA on regulating muscle K^+ , Na^+ homeostasis, the exercise-induced inactivation of NKA and their implications for fatigue.

MUSCLE IONIC CONCENTRATION CHANGES WITH CONTRACTIONS

To understand the potential roles of K^+ , Na^+ and Cl^- on muscle excitability and fatigue, it is essential to firstly identify their concentration changes in muscle interstitial and intracellular spaces during contractions. Changes in plasma ion concentrations, water shifts, and muscle ion contents with exercise have been reviewed elsewhere (20; 61; 86) and are beyond the scope of this review.

Muscle extra- and intracellular K^+ concentrations

Extracellular. Studies using a microdialysis technique to collect interstitial fluid directly from contracting human muscles, reveal that muscle interstitial $[K^+]$ ($[K^+]_{int}$) greatly exceeds the corresponding venous effluent $[K^+]$, by as much as 4-8 mM (36; 68; 89). Thus, venous $[K^+]$ cannot be used to predict $[K^+]_{int}$. The $[K^+]_{int}$ rises with increasing exercise intensity during sub-maximal and intense one-legged knee-extension exercise, reaching mean values as high as 11-13 mM during intense exhaustive exercise and with some individual values exceeding 15 mM (Fig. 1) (35; 36; 47; 68; 89).

Intracellular. As expected, intracellular $[K^+]$ ($[K^+]_i$) decreases substantially during muscular activity. In human muscle, $[K^+]_i$ declined from ~165 mM at rest to ~130 mM during intense exercise (87). Reductions in $[K^+]_i$ have been more extensively documented in animal models. Fast twitch muscles, like extensor digitorum longus (EDL), have greater resting $[K^+]_i$ than slow twitch muscles, like soleus (46; 66; 95). Moreover, continuous 20 Hz stimulation induced a greater decrease in $[K^+]_i$ in rat EDL (from 150 to 90 mM, a 1.7-fold decrease), than in soleus (from 130 to 100 mM, a 1.3-fold decrease) (66). The decline in $[K^+]_i$ with intermittent contractions is in the range of 1.2 to 1.4-fold (46). Similar (1.3- to 1.7)-fold decreases in $[K^+]_i$ were reported in stimulated amphibian muscles (7; 30).

Muscle extra- and intracellular Na⁺ concentrations

Extracellular. Small elevations in plasma [Na⁺] occur with exercise, which obscure an actual decline in plasma Na⁺ content, because the latter is less than the corresponding decline in plasma volume (61). A tendency to lowered interstitial [Na⁺] ([Na⁺]_{int}) from ~140 to ~130 mM has been reported during exhaustive incremental exercise, whilst venous [Na⁺] increased slightly (89). Thus, as observed for K⁺, venous [Na⁺] cannot be used as an estimate of [Na⁺]_{int}.

Intracellular. Muscle intracellular [Na⁺] ([Na⁺]_i) increases markedly during contractions. In humans, [Na⁺]_i rose from 6 mM at rest to 24 mM at the end of intense exercise, calculated from muscle Na⁺ content and inulin space determinations (87). Contrary to the situation observed with [K⁺]_i, fast twitch muscles have lower [Na⁺]_i (10-15 mM) than slow twitch muscles (14-25 mM) (25; 46; 66; 95). Electrical stimulation studies in animal muscles also revealed large increases in [Na⁺]_i. Fatiguing mouse soleus with tetanic contractions at a rate of 1 contraction per s for 5 min raised [Na⁺]_i from 14 mM to about 35 mM, representing a 1.5-fold increase above rest (25). Similarly, one tetanic contraction per s for 1 min raised [Na⁺]_i from 12 to 24 mM in mouse soleus muscle (46). Increases in [Na⁺]_i by as much as 2.5-fold have also been reported in rat EDL and frog sartorius muscles (30; 66).

Muscle extra- and intracellular Cl⁻ concentration

Extracellular. Plasma [Cl⁻] changes little during exercise in humans because of a concomitant loss of Cl⁻ and water (11; 49; 55; 57). We are unaware of any measures of muscle [Cl⁻]_{int} with exercise, but a decline can be anticipated, as observed for [Na⁺]_{int}.

Intracellular. Whilst muscle Cl⁻ content increases during fatiguing exercise (30; 58), increases in [Cl⁻]_i are less clear, in part due to a gain in intracellular water. After intense cycling exercise, [Cl⁻]_i was unchanged in human muscles (49). In a perfused rat hindlimb, stimulation to fatigue causing 40% force reduction, [Cl⁻]_i was unchanged in soleus and gastrocnemius but

almost doubled from 12 to 23 mM in plantaris (56). Muscle $[Cl^-]_i$ was increased after electrical stimulation in rat red gastrocnemius and plantaris muscles, but not in soleus or white gastrocnemius; exhaustive swimming did not increase $[Cl^-]_i$ in any of these muscles (58). Apparently, changes in $[Cl^-]_i$ and water uptake may vary significantly between muscles and activation patterns. Further studies investigating the effects of exercise upon muscle $[Cl^-]_i$ in humans are required.

DEPRESSING EFFECTS OF Na^+ , K^+ AND Cl^-

K^+

The resting E_m in unfatigued muscles ranges between -75 and -85 mV and decreases by ~10-15 mV when fatigue is elicited by intermittent contractions (7; 8; 21; 48; 54; 79; 80; 82; 93). Greater depolarizations of 25-30 mV occurred with prolonged and continuous stimulation of *Xenopus* toe muscle (51; 52). Two studies have reported E_m hyperpolarization with fatigue (39; 60); in both studies, the fatigue resistant soleus muscle was fatigued at 37°C. The fatigue-induced depolarization is most likely related to the increased $[K^+]_e$ and decreased $[K^+]_i$ which are known to cause membrane depolarization in unfatigued muscle (1; 15; 16; 23; 40; 81; 94). However, the depolarization observed during fatigue is considerably less than what is expected from the changes in $[K^+]_e$ and $[K^+]_i$. For example, doubling $[K^+]_{int}$ from 4 to 8 mM with a concomitant decrease in $[K^+]_i$ from 150 to 90 mM reduces the K^+ equilibrium potential (E_K) by 33 mV, and by 48 mV when $[K^+]_{int}$ reaches 14 mM as during intense contractions. Thus, whilst the dramatic changes in $[K^+]_{int}$ and $[K^+]_i$ that occur during fatigue most likely contribute to E_m depolarization, other ions, e.g., Cl^- (23), and the electrogenic effects of NKA activity (20) reduce the extent of K^+ -induced depolarization during fatigue. The main effect of membrane

depolarization is the inactivation of Na^+ channels (2; 41; 44; 45; 83), which lowers AP amplitude (81; 94; 95).

The dose-response curve for the K^+ effect on maximal tetanic force reveals a range of extracellular $[\text{K}^+]$ ($[\text{K}^+]_e$) causing little or no force depression despite the decrease in AP amplitude, followed by a precipitous decrease in force at higher $[\text{K}^+]_e$ (Fig. 2). Regardless of the species and experimental temperature, 90% of the decreases in twitch and tetanic force occur within 2-3 mM $[\text{K}^+]_e$ (12; 15; 16; 22; 75; 81; 94). The $[\text{K}^+]_e$ at which force abruptly declines, here referred as the critical $[\text{K}^+]_e$, is 9-10 mM in frog sartorius muscle (12; 81), whilst in *Xenopus* toe muscle, 14 mM $[\text{K}^+]_e$ still does not suppress twitch force (51). In mammalian muscles, the critical $[\text{K}^+]_e$ is 8-9 mM at 25-30°C (15; 16), but reaches 10-12 mM at 35-37°C (22; 75; 94). The abrupt K^+ -depressor effect is even more drastic if force is plotted against membrane potential, as the depression is completed within 5 mV, starting at -65 mV in frog sartorius (81) and at -60 mV in mouse soleus and EDL muscles (15; 16).

Na^+

Investigation of Na^+ effects on muscle contractility is more complex than K^+ , because the substantial $[\text{Na}^+]_i$ changes during fatigue are difficult to induce in unfatigued muscle without potentially complicating pharmacological interventions. However, if Na^+ effects occur via an effect on E_m , these can be mimicked by reducing $[\text{Na}^+]_e$ to replicate the fall in $[\text{Na}^+]_{\text{int}}/[\text{Na}^+]_i$ gradient observed during fatigue (13). For example, the decline in the $[\text{Na}^+]$ gradient due to 2- and 3-fold increases in $[\text{Na}^+]_i$, would be mimicked by experimentally reducing $[\text{Na}^+]_e$ from 150 mM to 75, and 50 mM, respectively.

Contrary to K^+ , Na^+ has little effects on resting E_m (1; 13; 40; 94). Instead, lower $[\text{Na}^+]_{\text{int}}/[\text{Na}^+]_i$ gradients reduce Na^+ influx during APs resulting in lower amplitudes (10; 13; 42).

Decreases in $[\text{Na}^+]_e$ or increases in $[\text{Na}^+]_i$ in unfatigued muscle reduce both twitch and tetanic force (10; 12; 13; 17; 59; 67). However, large force depressions only occurred when the $[\text{Na}^+]$ gradient is substantially reduced. For example, in mouse soleus muscle, at least 2.5-fold decrease in $[\text{Na}^+]_{\text{int}}/[\text{Na}^+]_i$ gradient was required to cause a 50% reduction in force (13). Furthermore, the decreases are observed within a very narrow change in $[\text{Na}^+]_e$, i.e., between 40 and 30 mM (13).

Cl⁻

Cl^- significantly contributes to the resting E_m (23; 40). In amphibian muscles, the equilibrium potential of Cl^- (E_{Cl}) is very close to the resting E_m and unlike changes in $[\text{K}^+]_e$, a change in $[\text{Cl}^-]_e$ only causes transient changes in resting E_m (40). In mammalian muscles at 37°C, E_{Cl} is closed to resting E_m in some muscles, such as rat diaphragm, while it is less negative in others, such as for rat red sternomastoid, mouse EDL and soleus muscle (23). During fatigue, any increase in $[\text{Cl}^-]_i$ is expected to lower E_{Cl} and to thus contribute to E_m depolarization. There are, however, very few studies on the effect of Cl^- on muscle contractility. One of these studies has demonstrated that lowering $[\text{Cl}^-]_e$ increased the rate of fatigue during continuous and intermittent contractions in rat and mouse soleus (14; 18).

PARADOXICAL EFFECTS OF K^+ , Na^+ AND H^+ ON CONTRACTILITY

Recent studies have demonstrated that changes in Na^+ , K^+ and H^+ not only have depressing effects but they can also enhance muscle contractility.

K⁺-elevation can potentiate muscle force

Although elevated $[\text{K}^+]_e$ that remains below the critical $[\text{K}^+]_e$ has no effect on the maximum force generated during a completely fused tetanus, it actually enhances forces generated with submaximal tetanic stimulation. In frog sartorius muscle, twitch potentiation reached a maximum of 60% at 8-9 mM (12; 81), whilst in *Xenopus* toe muscle, peak twitch force

was 2-fold higher at 14 mM compared to 2.5 mM K^+ (51). In mammalian muscles, twitch potentiation reaches a maximum at 9-10 mM $[K^+]_e$, being 15-20% at 20-25°C (16) and 80-100% at 37°C (43; 94). Similar twitch potentiation was observed for rat gastrocnemius muscle *in situ* at a plasma $[K^+]$ of 14-15 mM (92) and in rabbit hindlimb extensor muscles at 10 mM (37). The K^+ -induced force potentiation is also not limited to twitch contraction. Raising $[K^+]_e$ from 2 to 8 mM at 37°C, increases sub-tetanic peak force by 60% in soleus stimulated at 12 Hz and by 2-fold in EDL muscles stimulated at 40 Hz (43). In rabbit hindlimb extensor muscles, 10 mM $[K^+]$ potentiated sub-tetanic peak force up to 80 Hz, and reversed fatigue elicited with 0.25 Hz (37). Finally, force potentiation is 60-100% when the stimulation frequency is equal or less than 110 Hz in mouse EDL, and reached 2-3-fold in soleus at stimulation frequencies up to 40Hz (He and Renaud, unpublished results). Interestingly, the typical in-vivo motor unit discharge rates for rat soleus muscles are ~10-20 Hz, and 90-100 Hz for EDL muscles (38). Thus, the K^+ -induced sub-maximal tetanic force potentiation is within a physiological range of stimulation frequencies for each of soleus and EDL. The mechanism of the K^+ -induced force potentiation is still not understood, but at least in mouse EDL, it is not due to a prolongation of the AP allowing for longer Ca^{2+} release (94).

Attenuation of the K^+ -induced force depression by Na^+ , H^+ and lactic acid

In electrically-stimulated, mechanically skinned single fibers, the depressive effects of substantially lowering $[K^+]_i$ (from 120 to 20 mM) were reduced by elevated $[Na^+]_i$ (70). This was suggested to reflect effects of increased NKA activity on the t-tubular membrane (70). Thus, whilst a reduced $[Na^+]_{int}/[Na^+]_i$ gradient by reducing $[Na^+]_e$ causes force depression, elevated $[Na^+]_i$ can also attenuate this effect.

Lactic acid, at 20 mM, increases the critical $[K^+]_e$ at which force depression occurs, by almost 2 mM $[K^+]$ (22; 75) (Fig. 2), likely due to an increase in $[H^+]_i$ and not lactic acid per se (22). The higher force was associated with a recovery of the M-wave area suggesting that the $[H^+]_i$ effect involves a recovery of membrane excitability. In a subsequent study, it was shown that an acidic pH_i lowers Cl^- conductance allowing for greater AP amplitude at 11 mM $[K^+]$ and thus greater force development (76). An improved force development due to lower Cl^- conductance at 11 mM $[K^+]_e$ was further supported by findings that inhibiting Cl^- channels with 9-AC increased force at 11 mM $[K^+]$ (76). Finally, the force of mechanically skinned fibers, elicited by electrical stimulation of t-tubules, was less depressed by lowering $[K^+]_i$ in acidic than normal pH, an effect not observed when Cl^- was removed (77). Together, these studies demonstrated that when the AP is largely depressed at high $[K^+]_e$, a reduction in Cl^- conductance lowers the Cl^- influx during the AP allowing for greater Na^+ -induced depolarization. As the amplitude of the AP increases, the amount of Ca^{2+} release and thus force increases.

Na^+ , K^+ AND Cl^- EFFECTS: IMPLICATIONS FOR FATIGUE

For years, increases in lactic acid, increases in $[H^+]_i$ and changes in $[Na^+]_i$, $[K^+]_i$, $[K^+]_e$, and $[Cl^-]_i$ have been proposed as being factors contributing to the decrease in force during fatigue. Recent studies using unfatigued muscles, have demonstrated that some ions can alleviate fatigue effects, or even potentiate sub-maximal tetanic force. Thus the effects of Na^+ , K^+ and Cl^- on contractility are more complex than originally thought.

The effects observed in unfatigued muscles may reflect events occurring early in exercise. For example, during a 30-min knee-extensor exercise bout at 30 W, $[K^+]_{int}$ increases to 10 mM within 5 min and remained above 8 mM for the remaining exercise period (68). Any depressive effect of 10 mM $[K^+]$ may have been minimized by increased catecholamines and

$[H^+]_i$ (via reducing Cl^- conductance), NKA activation (via increased $[Na^+]_i$). The increased $[K^+]_{int}$ may also have enhanced muscle performance via potentiation of sub-maximal force and vasodilatation of blood vessels (3). Thus, the effects of K^+ , Na^+ , H^+ and Cl^- in unfatigued muscle may represent a mechanism that maximizes muscle performance early in exercise.

The effects observed in unfatigued muscles do not, however, eliminate perturbations in these ions as potential causative factors in the etiology of muscle fatigue. Numerous observations make it attractive to hypothesize that high $[K^+]_{int}$ per se contributes to fatigue development in human muscle. Firstly, the high $[K^+]_{int}$ (12-14 mM) reported in human muscle with fatigue is within the range of critical $[K^+]_e$, identified by in vitro studies, that caused marked force depression. Secondly, the decline in human vastus lateralis E_m from ~90 to ~75 mV during exhaustive exercise (87) might be an underestimate. When average values of $[K^+]_{int}$ and $[Na^+]_{int}$ at exhaustion are used rather than venous concentrations, E_m at exhaustion would be as low as ~60 mV, at which muscle fiber contractility is significantly impaired. This calculation does not however take into account the effects of changes in K^+ and Cl^- conductance, but these are unknown in exercising human muscles. Thirdly, when knee-extensor exercise is preceded by arm-exercise, muscle $[K^+]_{int}$ increases more rapidly and fatigue occurs earlier when compared to a control exercise without any arm-exercise (72). Fourthly, training increases fatigue resistance while reducing the rate of increase in $[K^+]_{int}$ (68).

Some observations do not fully support a role for elevated $[K^+]_{int}$ per se in fatigue. A marked range of $[K^+]_{int}$ has been reported at exhaustion, suggesting that increases in $[K^+]_{int}$ per se do not always depress muscle function and exercise performance (71). When intense exercise is repeated, exhaustion occurs within 5.5 min at $[K^+]_{int}$ of 11.5 mM during the 1st bout of exercise, but within 3.5 min during the 3rd bout at a lower $[K^+]_{int}$ of 9 mM (Fig. 1) (64). Considering the

force- $[K^+]_e$ relationship, $[K^+]_{int}$ reaches the critical $[K^+]_e$ near the end of fatigue, especially in the presence of lactic acid and catecholamines, suggesting that most of the decrease in force may not be related to changes in $[K^+]_{int}$ and $[K^+]_i$.

While there are factors that increase the critical $[K^+]_e$ for force depression, there are also factors that reduce the critical $[K^+]_e$, including Na^+ and Cl^- (Fig. 2). In rat soleus muscle (74), a 1.3-fold increase in $[K^+]_e$, or a 1.7-fold decrease in $[Na^+]_e$ from 147 to 85 mM reduces peak tetanic force by 10%. When $[K^+]_e$ and $[Na^+]_e$ are changed simultaneously, peak tetanic force decreased by 50%, suggesting a synergistic rather than an additive effect for Na^+ and K^+ . Furthermore, all reductions in peak tetanic force correlated well with reductions in M-waves area, suggesting that the Na^+ and K^+ depressor effects involved a decrease in membrane excitability. A similar Na^+ and K^+ synergistic effect was reported in frog sartorius muscle (12). Interestingly, both the 1.3-fold increase in $[K^+]_e$ and 1.7-fold reduction in $[Na^+]_e$ gradient are smaller than the changes observed during fatigue. Similarly, while lowering $[Cl^-]_e$ causes a transient decrease in force and 9 mM $[K^+]_e$ almost none in unfatigued muscle, concomitantly lowering $[Cl^-]_e$ and increasing $[K^+]_e$ results in large and synergistic decrease in force (18). It appears that $[K^+]_{int}$ together with changes in $[K^+]_i$, $[Na^+]_i$, and $[Cl^-]_i$ must be considered not separately, but in combination, to better understand their role in the etiology of muscle fatigue. Further, NKA activity modulates each of $[K^+]_i$, $[K^+]_{int}$ and $[Na^+]_i$ and E_m and thus must also impact on muscle force and fatigability.

Na^+, K^+ -ATPASE AND MUSCLE EXCITABILITY

Muscle contraction elicits a rapid increase in NKA Activity

Muscle excitation results in a dramatic increase in the NKA activity, as measured in-vitro by ^{22}Na extrusion or ^{86}Rb uptake (20). Even brief contractions of 2 seconds are sufficient to

markedly stimulate the NKA in isolated muscles, indicating that elevated NKA activity must occur at the commencement of all muscular exercise. In isolated rat soleus muscles subjected to high frequency electrical stimulation, NKA activity may reach the maximal theoretical activity of the pump, indicating that all available pumps are fully activated (20; 24; 69). It is likely however, that the extent of NKA activation is considerably less under physiological conditions. In contracting muscles in humans, evidence for in-vivo rates of NKA activation has been based on venous $[K^+]$ changes using an indwelling K^+ -sensitive microelectrode; these studies suggest that NKA activity increases to only 25-50% of the maximal theoretical activity (86). These calculations will however, likely be underestimates due to the very large (4-6 mM) interstitial-venous $[K^+]$ gradient that develops during exercise (36; 68). Regardless, a vigorous activation of NKA clearly occurs with exercise, evidenced by the rapid and pronounced decline in venous $[K^+]$ post-exercise (91). Most studies indicate increased $[Na^+]_i$ with exercise, which is a potent NKA stimulator; this suggests ongoing NKA activity regardless of exercise duration. Furthermore, NKA activity remains elevated even with declining intracellular $[Na^+]$ (69), with NKA activation related to events associated with membrane excitation. Thus, elevated NKA activity must be an important feature of all muscular contraction, which has important implications both for constraining fatigue through regulation of trans-sarcolemmal K^+ and Na^+ gradients and E_m , as well as exacerbating fatigue when at least some pumps become inactivated.

Measurement of NKA activity in human skeletal muscle

NKA activity in isolated muscles can be assessed by ouabain-inhibitable ^{22}Na transport or ^{86}Rb uptake, but such approaches are impossible in exercising humans in vivo. Thus in vitro approaches have been applied to muscle obtained via biopsy to determine NKA activity. Due to limited tissue yield the traditional measures of NKA activity, P_i liberation or ATP hydrolysis

have not been employed; rather an activity measure utilising the properties of a partial reaction of the NKA enzyme cycle have been utilised. This assay measures the phosphatase activity that is stimulated by K^+ and inhibitable by ouabain, and modified for human skeletal muscle (28). The phosphatase cleaves the artificial substrate 3-O-methyl fluorescein phosphate, yielding the fluorescent compound 3-O-methyl fluorescein and is referred to as the 3-O-methyl fluorescein phosphatase (3-O-MFPase) activity (28).

Exercise inhibits maximal NKA Activity in human muscle.

Work from two laboratories in recent years has investigated the effects of fatiguing exercise on the maximal NKA (in-vitro 3-O-MFPase) activity in human muscle. Findings have consistently revealed reduced maximal NKA activity after exhaustive exercise (Fig. 3), including after sustained submaximal isometric contractions (26); dynamic sprinting or quadriceps contractions lasting 1-6 minutes duration (4; 27; 78); repeated high intensity bouts (6); incremental exercise (5; 84); through to submaximal cycling from 70%-90% VO_{2peak} leading to fatigue in excess of one hour (53; 65). Furthermore, this depression has been demonstrated in untrained and highly trained athletes, and is repeatable in the same athlete, indicating a persistent effect (4; 27). There are two interesting exceptions. A recent report indicated elevated in-vitro NKA activity during exercise at 57% VO_{2peak} when participants undertook glucose feeding; this was attributed to an increased appearance of pumps due to translocation, evidenced by an accompanying increase in ouabain binding (34). Furthermore, no decline in maximal in-vitro NKA activity was observed during the placebo exercise trial (34). This seems unlikely to reflect a minimum level of exercise and accompanying intramuscular disturbance required for this effect to be manifest, since the same group reported a decline in activity at a similar relative work rate (84). We were recently unable to detect any decline in NKA activity in electrically stimulated,

isolated soleus or EDL muscle in the rat, despite a marked loss of force (31). The latter may reflect increased resistance to exercise-induced inactivation in the rodent, consistent with small inhibitory effects observed in running rats only after very prolonged exercise (26). Indeed an increased maximal NKA activity has even been reported in rats during prolonged running exercise (85).

The mechanisms underlying these inhibitory exercise effects on muscle NKA activity have not yet been extensively studied. Several studies have demonstrated that the total number of pumps in skeletal muscle, as fully quantified in human muscle by ouabain binding, is not reduced with exercise (4; 53; 65). Thus the decline in NKA activity is referred to as NKA inactivation rather than a loss of existing pumps. A recent study has implicated reactive oxygen species (ROS) in NKA inactivation with exercise (62). During prolonged submaximal exercise comprising 45 minutes at 70% $\text{VO}_{2\text{peak}}$ followed by exercise to exhaustion at 70%-90% $\text{VO}_{2\text{peak}}$, maximal NKA activity underwent a substantial decline. Infusion of the non-specific antioxidant *N*-acetylcysteine (NAC) prolonged the time to fatigue and almost halved the relative decline in NKA activity (62). Thus redox disturbances may be responsible for at least an important fraction of NKA inactivation. This observation is also consistent with the important role for reactive oxygen species in muscle fatigue (Reid review in this series). Another potential mechanism is related to the pulsatile increases in $[\text{Ca}^{2+}]_i$ during muscle contraction. Incubation of diaphragm in high extracellular Ca^{2+} concentration, which increased $[\text{Ca}^{2+}]_i$, reduced NKA activity (90). Whether such effects occur in a physiological preparation, or in vivo remain to be seen.

The modest magnitude of these depressive effects might suggest that NKA inactivation has little functional adverse effect. However, these small changes are consistent with similar small magnitude of upregulatory changes seen with training in NKA activity (6; 32), NKA

content (33; 63) or NKA isoform protein abundance (32; 68) Furthermore, skeletal muscle NKA activity may be regulated independently of NKA content with training (6; 32).

Implications of NKA inactivation for fatigue: muscle inexcitability with exercise

The implications of NKA inactivation for fatigue are important, though not yet resolved. Declining NKA activity has the potential to impair each of the rates of Na^+ extrusion, and K^+ uptake, as well as the electrogenic effect on E_m . These changes could exacerbate localised increases in $[\text{Na}^+]_i$ and in $[\text{K}^+]_{int}$, reductions in $[\text{K}^+]_i$ and in E_m , culminating in a reduction in muscle force/power, via mechanisms described above. Important unresolved issues include whether the reduction in maximal NKA activity reflects a severe loss of activity in some fibres, or a more generalised modest decrease in activity in many fibres. The implications would be quite different for the two scenarios: severe NKA inactivation could induce severe local cellular ionic disruptions and lead to complete loss of excitability in those fibres. A more generalised lesser decline would have only smaller effects per se on muscle ionic homeostasis and thus also on fatigue. Further work is also required to determine whether functional inactivation occurs in vivo during exercise, as present results of this phenomenon are all collected in vitro.

Evidence linking muscle NKA inactivation with membrane inexcitability in exercising humans has not yet been extensively studied. Studies from the same laboratory combining both NKA activity and M wave characteristics produced inconsistent results: i) both M wave area and NKA activity were reduced after isometric contractions (26); ii) NKA activity was reduced but M wave area unchanged after exhaustive incremental exercise (84); and iii) NKA activity was increased but M-wave area reduced after prolonged exercise (34; 88). Further research is clearly required to resolve this important issue.

UNDERSTANDING THE ROLES OF K^+ , Na^+ AND Cl^- : FUTURE DIRECTIONS

If the enhancing and depressing effects of K^+ , Na^+ , H^+ and Cl^- are all physiologically relevant, future research needs to identify the important local conditions relevant to the opposing

effects, and their underlying mechanisms. Evidence exists for a mitochondrial factor that is released when ATP production is incapacitated and that reduces t-tubular membrane excitability (73). The concept that fatigue is related at least in part to a failure of membrane excitability has important implications not only for understanding factors limiting muscle force/power, but also for cellular Ca^{2+} homeostasis and energy metabolism. Muscle fatigue acts to preserve cellular integrity via regulation of $[\text{Ca}^{2+}]_i$ and ATP levels, and thus survival. At least three membrane components are sensitive to either ATP or Ca^{2+} : i) the ATP-sensitive K^+ channel (K_{ATP} channel), ii) the CLC-1 Cl^- channel (9), and iii) the Ca^{2+} -sensitive K^+ channel (50). An increased activity of any of these three components undoubtedly will contribute to fatigue. This has already been shown for the K_{ATP} channels which allow greater K^+ efflux and directly lower AP amplitude (19; 29; 60). In regard to the Cl^- channel, an increased activity will also lower AP amplitude as Cl^- influx increases. It will therefore be interesting to determine whether the mitochondrial factor activates any of these channels. A final potential target is NKA because an inactivation of the pump would exacerbate increases in $[\text{K}^+]_{\text{int}}$ and $[\text{Na}^+]_i$ which then would favor the depressing over enhancing effects. Further studies are required to quantify the extent of in-vivo NKA activity with exercise in human muscles, and to determine the combined effects of increased activity above rest levels plus NKA inactivation on muscle excitability and fatigue effects.

CONCLUSIONS

In conclusion, marked perturbations in each of $[\text{Na}^+]_i$, $[\text{K}^+]_i$, and $[\text{K}^+]_{\text{int}}$ occur during fatiguing exercise in humans and during electrical stimulation in in-vitro preparations, with less well established changes in $[\text{Na}^+]_{\text{int}}$, $[\text{Cl}^-]_i$ and $[\text{Cl}^-]_{\text{int}}$. In unfatigued muscle, these ionic changes depress force, but under some conditions they can also enhance force or alleviate the fatigue effect; i.e., the effects of these ions are more complex than originally thought. It is proposed that

these enhancing or alleviating effects are important early in exercise while their depressing effects are dominant later and directly contribute to development of muscle fatigue. Similarly, exercise results in an early and sustained increase in NKA activity, which potentiates contractility, whereas later in exercise, or after intense contractions a decline in maximal NKA activity occurs and contributes to fatigue.

REFERENCE LIST

1. **Adrian RH.** The effect of internal and external potassium concentration on the membrane potential of frog muscle. *J Physiol (Lond)* 133: 631-658, 1956.
2. **Adrian RH, Chandler WK and Hodgkin AL.** Voltage clamp experiments in striated muscle fibres. *J Physiol (Lond)* 208: 607-644, 1970.
3. **Armstrong ML, Dua AK and Murrant CL.** Potassium initiates vasodilatation induced by a single skeletal muscle contraction in hamster cremaster muscle. *J Physiol (Lond)* 581: 841-852, 2007.
4. **Aughey RJ, Clark SA, Gore CJ, Townsend NE, Hahn AG, Kinsman TA, Goodman C, Chow CM, Martin DT, Hawley JA and McKenna MJ.** Interspersed normoxia during live high, train low interventions reverses an early reduction in muscle Na^+ , K^+ ATPase activity in well-trained athletes. *Eur J Appl Physiol* 98: 299-309, 2006.
5. **Aughey RJ, Gore CJ, Hahn AG, Garnham AP, Clark SA, Petersen AC, Roberts AD and McKenna MJ.** Chronic intermittent hypoxia and incremental cycling exercise independently depress muscle in-vitro maximal Na^+ , K^+ ATPase activity in well-trained athletes. *J Appl Physiol* 98: 186-192, 2005.
6. **Aughey RJ, Murphy KT, Clark SA, Clark SA, Garnham AP, Snow RJ, Cameron-Smith D, Hawley JA and McKenna MJ.** Muscle Na^+ - K^+ -ATPase activity and isoform adaptations to intense interval exercise and training in well-trained athletes. *J Appl Physiol* 103: 39-47, 2007.
7. **Balog EM and Fitts RH.** Effects of fatiguing stimulation on intracellular Na^+ and K^+ in frog muscle. *J Appl Physiol* 81: 679-685, 1996.

8. **Balog EM, Thompson LV and Fitts RH.** Role of sarcolemma action potentials and excitability in muscle fatigue. *J Appl Physiol* 76: 2157-2162, 1994.
9. **Bennetts B, Rychkov GY, Ng H-L, Morton CJ, Stapleton D, Parker MW and Cromer BA.** Cytoplasmic ATP-sensing domains regulate gating of skeletal muscle ClC-1 chloride channels. *J Biol Chem* 280: 32452-32458, 2005.
10. **Bezanilla F, Caputo C, Gonzales-Serratos H and Venosa RA.** Sodium dependence of the inward spread of activation in isolated twitch muscle fibres of the frogs. *J Physiol (Lond)* 223: 507-523, 1972.
11. **Böning D, Klarholz C, Himmelsbach B, Hütler M and Maassen N.** Causes of differences in exercise-induced changes of base excess and blood lactate. *Eur J Appl Physiol* 99: 163-171, 2007.
12. **Bouclin R, Charbonneau E and Renaud JM.** Na⁺ and K⁺ effect on contractility of frog sartorius muscle: implication for the mechanism of fatigue. *Am J Physiol* 268: C1528-C1536, 1995.
13. **Cairns SP, Buller SJ, Loiselle DS and Renaud JM.** Changes of action potentials and force at lowered [Na⁺]_o in mouse skeletal muscle: implication for fatigue. *Am J Physiol Cell Physiol* 285: C1131-C1141, 2003.
14. **Cairns SP and Dulhunty AF.** High-frequency fatigue in rat skeletal muscle: role of extracellular ion concentrations. *Muscle Nerve* 18: 890-898, 1995.
15. **Cairns SP, Flatman JA and Clausen T.** Relation between extracellular [K⁺], membrane potential and contraction in rat soleus muscle: modulation by the Na⁺-K⁺ pump. *Pflugers Arch* 430: 909-915, 1995.

16. **Cairns SP, Hing WA, Slack JR, Mills RG and Loiselle DS.** Different effects of raised $[K^+]_o$ on membrane potential and contraction in mouse fast- and slow-twitch muscle. *Am J Physiol* 273: C598-C611, 1997.
17. **Cairns SP, Hing WA, Slack JR, Mills RG and Loiselle DS.** Role of extracellular $[Ca^{2+}]$ in fatigue of isolated mammalian skeletal muscle. *J Appl Physiol* 84: 1395-1406, 1998.
18. **Cairns SP, Ruzhynsky V and Renaud JM.** Protective role of extracellular chloride in fatigue of isolated mammalian skeletal muscle. *Am J Physiol* 287: C762-C770, 2004.
19. **Cifelli C, Bourassa F, Gariépy L, Banas K, Benkhalti M and Renaud J.M.** K_{ATP} channel deficiency in mouse FDB causes fiber damage and impairs Ca^{2+} release and force development during fatigue in vitro . *J Physiol (Lond)* 582: 843-857, 2007.
20. **Clausen T.** Na^+-K^+ pump regulation and skeletal muscle contractility. *Physiol Rev* 83: 1269-1324, 2003.
21. **Comtois AS, Light PE and Renaud JM.** Effect of tolbutamide on the rate of fatigue and recovery in frog sartorius muscle. *J Pharmacol Exp Ther* 274: 1061-1066, 1995.
22. **de Paoli FV, Overgaard K, Pedersen TH and Nielsen OB.** Additive protective effects of the addition of lactic acid and adrenaline on excitability and force in isolated rat skeletal muscle depressed by elevated extracellular K^+ . *J Physiol (Lond)* 581: 829-839, 2007.
23. **Dulhunty AF.** The dependence of membrane potential on extracellular chloride concentration in mammalian skeletal muscle fibres. *J Physiol (Lond)* 276: 67-82, 1978.
24. **Everts ME and Clausen T.** Excitation-induced activation of the Na^+-K^+ pump in rat skeletal muscle. *Am J Physiol* 266: C925-C934, 1994.

25. **Fong CN, Atwood HL and Charlton MP.** Intracellular sodium activity at rest and after tetanic stimulation in muscles of normal and dystrophic ($dy^{2j/dy2j}$)C57Bl/6J mice. *Exp Neurol* 93: 359-368, 1986.
26. **Fowles JR, Green HJ, Tupling R, O'Brien S and Roy BD.** Human neuromuscular fatigue is associated with altered Na^+ - K^+ -ATPase activity following isometric exercise. *J Appl Physiol* 92: 1585-1593, 2002.
27. **Fraser SF, Li JL, Carey MF, Wang XN, Sangkabutra T, Sostaric S, Selig SE, Kjeldsen K and McKenna MJ.** Fatigue depresses maximal in vitro skeletal muscle Na^+ , K^+ -ATPase activity in untrained and trained individuals. *J Appl Physiol* 93: 1650-1659, 2002.
28. **Fraser SF and McKenna MJ.** Measurement of Na^+ , K^+ -ATPase activity in human skeletal muscle. *Anal Biochem* 258: 63-67, 1998.
29. **Gong B, Legault D, Miki T, Seino S and Renaud JM.** K_{ATP} channels depress force by reducing action potential amplitude in mouse EDL and soleus. *Am J Physiol Cell Physiol* 285: C1464-C1474, 2003.
30. **Gonzales-Serratos H, Somlyo AV, McClellan G, Shuman H, Borrero LM and Somlyo AP.** Composition of vacuoles and sarcoplasmic reticulum in fatigued muscle: electron probe analysis. *Proc Natl Acad Sci* 75: 1329-1333, 1978.
31. **Goodman, C. A. and Hayes, A.** Rat skeletal muscle 3-O-MFPase activity is not decreased by fatiguing in vitro electrical stimulation. Second biennial Conference on Potassium, Sodium and the Function of Heart and Skeletal Muscle , S7.2. 2007.

32. **Green HJ, Barr DJ, Fowles JR, Sandiford SD and Ouyang J.** Malleability of human skeletal muscle $\text{Na}^+\text{-K}^+\text{-ATPase}$ pump with short-term training. *J Appl Physiol* 97: 143-148, 2004.
33. **Green HJ, Chin ER, Ball-Burnett M and Ranney D.** Increases in human skeletal muscle $\text{Na}^+\text{-K}^+\text{-ATPase}$ concentration with short-term training. *Am J Physiol* 264: C1538-C1541, 1993.
34. **Green HJ and Duhamel TA.** Glucose supplements increase human muscle in vitro $\text{Na}^+\text{-K}^+\text{-ATPase}$ activity during prolonged exercise. *Am J Physiol* 293: R354-R362, 2007.
35. **Green S, Bülow J and Saltin B.** Microdialysis and the measurement of muscle interstitial K^+ during rest and exercise in humans. *J Appl Physiol* 87: 460-464, 1999.
36. **Green S, Langberg H, Skovgaard D, Bulow J and Kjar M.** Interstitial and arterial-venous $[\text{K}^+]$ in human calf muscle during dynamic exercise: effect of ischaemia and relation to muscle pain. *J Physiol (Lond)* 529: 849-861, 2000.
37. **Gutierrez G, Kiiski R, Fernadez E and Lee DH.** Reversal of muscle fatigue in intact rabbits by intravenous potassium chloride. *Journal of Critical care* 11: 197-205, 1996.
38. **Hennig R and Lomo T.** Firing patterns of motor units in normal rats. *Nature* 314: 164-166, 1985.
39. **Hicks A and McComas AJ.** Increased sodium pump activity following repetitive stimulation of rat soleus muscles. *J Physiol (Lond)* 414: 337-349, 1989.
40. **Hodgkin AL and Horowicz P.** The influence of potassium and chloride ions on the membrane potential of single muscle fibres. *J Physiol (Lond)* 148: 127-160, 1959.
41. **Hodgkin AL and Huxley AF.** A quantitative description of membrane current and its application to conduction and excitation in nerve. *J Physiol (Lond)* 117: 500-544, 1952.

42. **Hodgkin AL and Katz B.** The effect of sodium ions on the electrical activity of the giant axon of the squid. *J Physiol (Lond)* 108: 37-77, 1949.
43. **Holmberg E and Waldeck B.** On the possible role of potassium ions in the action of terbutaline on skeletal muscle contractions. *Acta Pharmacol Toxicol* 46: 141-149, 1980.
44. **Ildelfonse M and Rougier O.** Voltage-clamp analysis of the early current in frog skeletal muscle fibre using the double sucrose-gap method. *J Physiol (Lond)* 222: 373-395, 1972.
45. **Ildelfonse M and Roy G.** Kinetic properties of the sodium current in striated muscle fibres on the basis of the Hodgkin-Huxley theory. *J Physiol (Lond)* 227: 419-431, 1972.
46. **Juel C.** Potassium and sodium shifts during *in vitro* isometric muscle contraction, and the time course of the ion-gradient recovery. *Pflugers Arch* 406: 458-463, 1986.
47. **Juel C, Pilegaard H, Nielsen JJ and Bangsbo J.** Interstitial K⁺ in human skeletal muscle during and after dynamic graded exercise determined by microdialysis. *Am J Physiol* 278: R400-R406, 2000.
48. **Karelis AD, Péronnet F and Gardiner PF.** Resting membrane potential of rat plantaris muscle fibers after prolonged stimulation in situ: effect of glucose infusion. *Can J Appl Physiol* 30: 105-112, 2005.
49. **Kowalchuk JM, Heigenhauser GJF, Lindinger MI, Sutton JR and Jones NL.** Factors influencing hydrogen ion concentration in muscle after intense exercise. *J Appl Physiol* 65: 2080-2089, 1988.
50. **Kristensen M, Hansen T and Juel C.** Membrane proteins involved in potassium shifts during muscle activity and fatigue. *Am J Physiol* 290: R766-R772, 2004.

51. **Lännergren J and Westerblad H.** Force and membrane potential during and after fatiguing, continuous high-frequency stimulation of single *Xenopus* muscle fibres. *Acta Physiol Scand* 128: 359-368, 1986.
52. **Lännergren J and Westerblad H.** Action potential fatigue in single skeletal muscle fibres of *Xenopus*. *Acta Physiol Scand* 129: 311-318, 1987.
53. **Leppik JA, Aughey RJ, Medved I, Fairweather I, Carey MF and McKenna MJ.** Prolonged exercise to fatigue in humans impairs skeletal muscle Na^+ , K^+ -ATPase activity, sarcoplasmic reticulum Ca^{2+} release and Ca^{2+} uptake. *J Appl Physiol* 97: 1414-1423, 2004.
54. **Light PE, Comtois AS and Renaud JM.** The effect of glibenclamide on frog skeletal muscle: evidence for K^+ _{ATP} channel activation during fatigue. *J Physiol (Lond)* 475: 495-507, 1994.
55. **Lindinger MI, Graham TE and Spriet LL.** Caffeine attenuates the exercise-induced increase in plasma $[\text{K}^+]$ in humans. *J Appl Physiol* 74: 1149-1155, 1993.
56. **Lindinger MI and Heigenhauser GJF.** Ion fluxes during tetanic stimulation in isolated perfused rat hindlimb. *Am J Physiol* 254: R117, 1988.
57. **Lindinger MI, Heigenhauser GJF, McKelvie RS and Jones NL.** Blood ion regulation during repeated maximal exercise and recovery in humans. *Am J Physiol* 262: R126-R136, 1992.
58. **Lindinger MI, Heigenhauser GJF and Spriet LL.** Effects of intense swimming and tetanic electrical stimulation on skeletal muscle ions and metabolites. *J Appl Physiol* 63: 2331-2339, 1987.

59. **Macdonald WA, Nielsen OB and Clausen T.** $\text{Na}^+\text{-K}^+$ pump stimulation restores carbacholine-induced loss of excitability and contractility in rat skeletal muscle. *J Physiol (Lond)* 563: 459-469, 2005.
60. **Matar W, Nosek TM, Wong D and Renaud JM.** Pinacidil suppresses contractility and preserves energy but glibenclamide has no effect during fatigue in skeletal muscle. *Am J Physiol* 278: C404-C416, 2000.
61. **McKenna MJ.** The roles of ionic processes in muscular fatigue during intense exercise. *Sports Med* 13: 134-145, 1992.
62. **McKenna MJ, Medved I, Goodman C, Brown MJ, Bjorksten AR, Murphy KT, Petersen AC, Sostaric S and Gong X.** N-acetylcysteine attenuates the decline in muscle $\text{Na}^+\text{,K}^+$ -pump activity and delays fatigue during prolonged exercise in humans. *J Physiol (Lond)* 576: 279-288, 2006.
63. **McKenna MJ, Schmidt TA, Hargreaves M, Cameron L, Skinner SL and Kjeldsen K.** Sprint training increases human skeletal muscle $\text{Na}^+\text{-K}^+$ -ATPase concentration and improves K^+ regulation. *J Appl Physiol* 75: 173-180, 1993.
64. **Mohr M, Nordsbork N, Nielsen JJ, Pedersen LD, Fischer C, Krustup P and Bangsbo J.** Potassium kinetics in human muscle interstitium during repeated intense exercise in relation to fatigue. *Pflugers Arch* 448: 452-456, 2004.
65. **Murphy KT and Petersen AC.** Prolonged submaximal exercise induces isoform-specific $\text{Na}^+\text{-K}^+$ -ATPase mRNA and protein responses in human skeletal muscle. *Am J Physiol* 290: R414-424, 2006.

66. **Nagaoka R, Yamashita S, Mizuno M and Akaike N.** Intracellular Na^+ and K^+ shifts induced by contractile activities of rat skeletal muscles. *Comp Biochem Physiol [A]* 109: 957-965, 1994.
67. **Nakajima S, Nakajima Y and Bastian J.** Effects of sudden changes in external sodium concentration on twitch tension in isolated muscle fibers. *J Gen Physiol* 65: 459-482, 1975.
68. **Nielsen JJ, Mohr M, Klarskov C, Kristensen M, Krustup P and Juel C.** Effects of high-intensity intermittent training on potassium kinetics and performance in human skeletal muscle. *J Physiol (Lond)* 554: 857-870, 2004.
69. **Nielsen OB and Clausen T.** Regulation of Na^+ - K^+ pump activity in contracting rat muscle. *J Physiol (Lond)* 503: 571-581, 1997.
70. **Nielsen OB, Örtenblad N, Lamb GD and Stephenson DG.** Excitability of the t-tubular system in rat skeletal muscle: roles of K^+ and Na^+ gradients and Na^+ - K^+ pump activity. *J Physiol (Lond)* 557: 133-146, 2004.
71. **Nordsborg N, Mohr M, Pedersen LD, Nielsen JJ, Langberg H and Bangsbo J.** Muscle interstitial potassium kinetics during intense exhaustive exercise: effect of previous arm exercise. *Am J Physiol* 285: R143-R148, 2003.
72. **Nordsborg N, Mohr M, Pedersen LD, Nielsen JJ, Langberg H and Bangsbo J.** Muscle interstitial potassium kinetics during intense exhaustive exercise: effect of previous arm exercise. *Am J Physiol* 285: R143-R148, 2003.
73. **Örtenblad N and Stephenson DG.** A novel signaling pathway originating in mitochondria modulates rat skeletal muscle membrane excitability. *J Physiol (Lond)* 548: 139-145, 2003.

74. **Overgaard K, Nielsen OB, Flatman JA and Clausen T.** Relations between excitability and contractility in rat soleus muscle: role of the $\text{Na}^+\text{-K}^+$ pump and Na^+/K^+ gradients. *J Physiol (Lond)* 518: 215-225, 1999.
75. **Pedersen TH, Clausen T and Nielsen OB.** Loss of force induced by high extracellular $[\text{K}^+]$ in rat muscle: effect of temperature, lactic acid and β_2 -agonist. *J Physiol (Lond)* 551: 277-286, 2003.
76. **Pedersen TH, de Paoli F and Nielsen OB.** Increased excitability of acidified skeletal muscle: role of chloride conductance. *J Gen Physiol* 125: 237-246, 2005.
77. **Pedersen TH, Nielsen OB, Lamb GD and Stephenson DG.** Intracellular acidosis enhances the excitability of working muscle. *Science* 305: 1144-1147, 2004.
78. **Petersen AC, Murphy KT, Snow RJ, Leppik JA, Aughey RJ, Garnham AP, Cameron-Smith D and McKenna MJ.** Depressed $\text{Na}^+\text{-K}^+\text{-ATPase}$ activity in skeletal muscle at fatigue is correlated with increased $\text{Na}^+\text{-K}^+\text{-ATPase}$ mRNA expression following intense exercise. *Am J Physiol* 289: R266-R274, 2005.
79. **Renaud JM.** The effect of lactate on intracellular pH and force recovery of fatigued sartorius muscle of the frog, *Rana pipiens*. *J Physiol (Lond)* 416: 31-47, 1989.
80. **Renaud JM and Comtois A.** The effect of K^+ on the recovery of the twitch and tetanic force following fatigue in sartorius muscle of the frog, *Rana pipiens*. *J Mus Res Cell Mot* 15: 420-431, 1994.
81. **Renaud JM and Light P.** Effects of K^+ on the twitch and tetanic contraction in the sartorius muscle of the frog, *Rana pipiens*. Implication for fatigue *in vivo*. *Can J Physiol Pharmacol* 70: 1236-1246, 1992.

82. **Renaud JM and Mainwood GW.** The effects of pH on the kinetics of fatigue and recovery in frog sartorius muscle. *Can J Physiol Pharmacol* 63: 1435-1443, 1985.
83. **Ruff RL.** Effects of temperature on slow and fast inactivation of rat skeletal muscle Na⁺ channels. *Am J Physiol* 277: C937-C947, 1999.
84. **Sandiford SD, Green HJ, Duhamel TA, Schertzer JD, Perco JD and Ouyang J.** Muscle Na-K-pump and fatigue responses to progressive exercise in normoxia and hypoxia. *Am J Physiol* 289: R441-R449, 2005.
85. **Sandiford SDE, Green HJ and Ouyang J.** Mechanisms underlying increases in rat soleus Na⁺-K⁺-ATPase activity by induced contractions. *J Appl Physiol* 99: 2222-2232, 2005.
86. **Sejersted OM and Sjøgaard G.** Dynamics and consequences of potassium shifts in skeletal muscle and heart during exercise. *Physiol Rev* 80: 1411-1481, 2000.
87. **Sjøgaard G, Adams RP and Saltin B.** Water and ion shifts in skeletal muscle of humans with intense dynamic knee extension. *Am J Physiol* 248: R190-R196, 1985.
88. **Stewart RD, Duhamel TA, Foley JM, Ouyang J, Smith IC and Green HJ.** Protection of muscle membrane excitability during prolonged cycle exercise with glucose supplementation. *J Appl Physiol* 103: 331-339, 2007.
89. **Street D, Nielsen JJ, Bangsbo J and Juel C.** Metabolic alkalosis reduces exercise-induced acidosis and potassium accumulation in human skeletal muscle interstitium. *J Physiol (Lond)* 566: 481-489, 2005.
90. **Sulova Z and Vyskocil F.** Ca⁽²⁺⁾-induced inhibition of sodium pump: effects on energetic metabolism of mouse diaphragm tissue. *Gen Physiol Biophys* 17: 271-283, 1998.

91. **Vollestad NK, Hallen J and Sejersted OM.** Effect of exercise intensity on potassium balance in muscle and blood of man. *J Physiol (Lond)* 475: 359-368, 1994.
92. **Walker SM.** Action potentials in rat muscle with twitch tension potentiated by KCl treatment, adrenalectomy, tetanus and treppe. *Am J Physiol* 154: 63-72, 1948.
93. **Westerblad H and Lännergren J.** Force and membrane potential during and after fatiguing, intermittent tetanic stimulation of single *Xenopus* muscle fibres. *Acta Physiol Scand* 128: 369-378, 1986.
94. **Yensen C, Matar W and Renaud JM.** The K⁺-induced twitch potentiation is not due to longer action potential. *Am J Physiol* 283: C169-C177, 2002.
95. **Yonemura K.** Resting and action potentials in red and white muscles of the rat. *Jap J Physiol* 17: 708-719, 1967.

FIGURE 1

Extracellular (interstitial) $[K^+]$ in skeletal muscle during repeated intense exhaustive exercise in humans. $[K^+]_{int}$ was measured during three intense, exhaustive, knee-extensor exercise bouts (EX1, EX2 and EX3) separated by 10-min recovery periods (n=10). Means $\pm$ SD; *P<0.05 vs. EX1 at the point of exhaustion. Data from Mohr et al. (64).

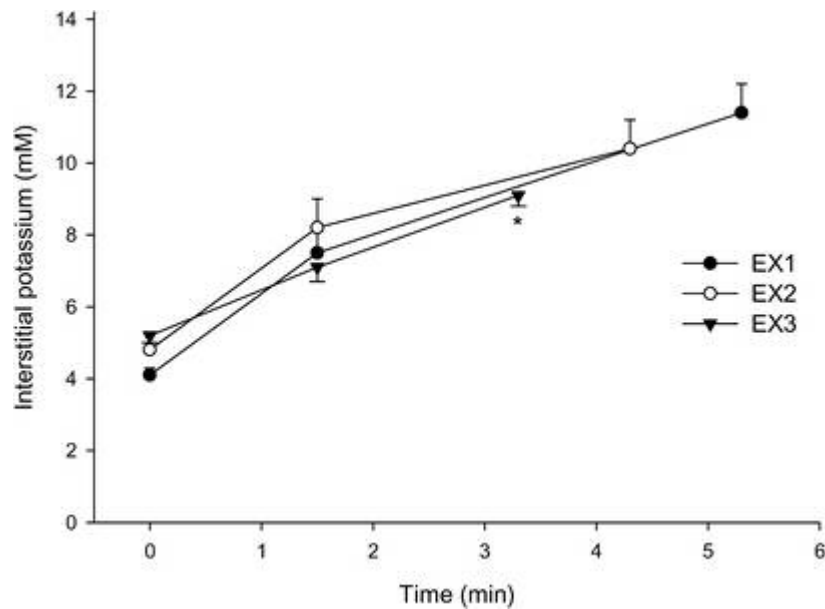


FIGURE 2

The peak tetanic force- $[K^+]_{int}$ relationship in skeletal muscle, indicating the critical $[K^+]_{int}$, the precipitous decline in force, and modulation of the relationship by other ions and by Na^+, K^+ -ATPase. The critical $[K^+]_{int}$ is defined as the interstitial $[K^+]$ at which peak tetanic force starts to decreased abruptly.

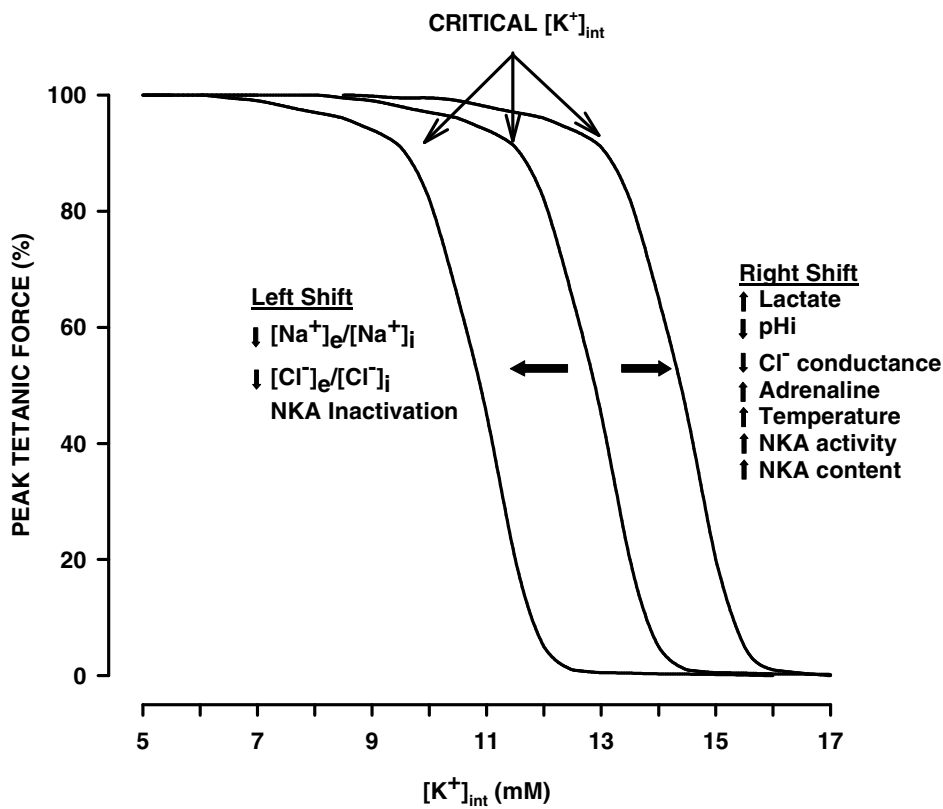


FIGURE 3

Depressive effects of exercise on maximal NKA activity in human skeletal muscle. NKA activity measured via in vitro maximal 3-O-MFPase activity and expressed as percentage change from rest (Pre-exercise). Exercise duration indicated on X-axis.

